Fiscal Year:	FY 2006	**Task Last Updated:**	FY 06/03/2007
PI Name:	Takahashi, Fumiaki D.Eng.		
Project Title:	Flame Stability, Spread, and Suppression		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Gaseous diffusion flames		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	fxt13@case.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	440-520-5939
Organization Name:	Case Western Reserve University		
PI Address 1:	Mechanical and Aerospace Engineering		
PI Address 2:	10900 Euclid Ave		
PI Web Page:			
City:	Cleveland	**State:**	OH
Zip Code:	44106-1712	**Congressional District:**	11
Comments:	NOTE: Also affiliated with NASA Glenn Research Center		
Project Type:	Ground	**Solicitation / Funding Source:**	2002 Physical Sciences 02-OBPR-03
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No. of Post Docs:	0	**No. of PhD Degrees:**	0
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No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	Note: Original end date was 12/30/2007.		
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):	Katta, Viswanath (Innovative Scientific Solutions, Inc.)		
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	This research relates to technology development efforts of both fire-extinguishing systems in human exploration of space and various combustion systems for terrestrial applications. The practical considerations are based on the fact that most uncontrolled fires and combustion systems are diffusion flame type combustion. Fire safety issues include flame stability, spread, and suppression phenomena aboard spacecraft in microgravity (ug) and on planetary surface bases in low gravity (the moon, 0.165g; Mars, 0.380g) as well as on the Earth (1g). As a result of flame-surface interactions, the diffusion flame often forms an edge, which may attach to burner walls, spread over condensed fuel surfaces, jump to another location through the fuel-air mixture formed, or extinguish by destabilization (blow-off). Furthermore, local extinction and re-ignition processes can occur in both fires and combustors as a result of interactions between the flame zone and vortices or fire-extinguishing agents. Flame spreading and secondary fires, where fires jump from one location to another, will lead an initially small ignition source to a		

	dangerous established fire. In such circumstances, the dynamic behavior of edge diffusion flames may play a key role in a fate of fires. Because of various levels of buoyancy in low-gravity, fires in space behave differently from those on Earth. This research at NASA Glenn addresses edge diffusion flame stability, spread, and suppression, which is of essential importance in space-based fire safety and Earth-bound combustion systems. The structure of edge diffusion flames, stabilized on various geometries are studied experimentally and computationally. The experiments include both 1g and low-g tests in the 2.2 s Drop Tower at Glenn and the C-9 Reduced-Gravity Aircraft. Diagnostics include video observations, the Mach-Zehnder Interferometry (MZI) for temperature measurements, and the particle image velocimetry (PIV) for velocity measurements. The fundamental aspect of the proposed program relates to recent advancement in numerical simulations. As a result of significant progresses in the development of detailed combustion reaction mechanisms and computer technologies over the last decade or two, it is now feasible to simulate various transient combustion phenomena in simple configurations (burner geometry, flow, and fuel) with confidence, leading to deeper understanding of physical and chemical unit processes taking place during the phenomena under investigation. Direct numerical simulations are performed by solving the time-dependent full Navier-Stokes equations along with energy- and species-conservation equations, including a detailed reaction mechanism for C1-C3 hydrocarbon combustion (33 species, 112 elementary steps) and radiative heat losses from CO2, H2O, CO, and CH4.
Task Description:	The objectives of this research are: 1. To elucidate flame stability, spread, and suppression mechanisms of edge diffusion flames of gaseous, liquid, and solid fuels using various fire-extinguishing agents in 1g and low-g by revealing internal flame structure. The edge diffusion flame geometries to be used include axisymmetric fuel jets, candles, and cylindrical fuel surfaces in a low-speed coaxial oxidant stream in 1g and low-g with the oxygen concentration, oxidant velocity, pressure, fuel type, and agent type as independent variables. 2. To provide rigorous testing of the fidelity of the computational models, including detailed chemical reaction mechanisms and the radiative heat-transfer sub-models, in the simulations of the subject phenomena: unsteady flame stabilization, spread, and suppression, for various edge diffusion flame geometries, environments, fuels, and agents. This research is important in relation to fire safety in space exploration. The diffusion flame stability mechanism is more complicated and less understood than that of premixed flames because the reaction rates in diffusion flames are transport-limited and complex flame-structure dependent. A conventional view of diffusion flame stability in normal gravity, where a buoyancy-driven flow exists, or for forced-convection conditions assumes that there exists a small premixed flame at the flame base, which propagates against the mixture flow, thus preventing the flame from lifting. This research thus far has revealed that the speculation of premixed flame base formation is unrealistic, particularly in diffusion dominant micro gravity flames, and a newly conceived concept based on the reaction kernel replaces the conventional view. Direct numerical simulations with a detailed reaction mechanism and radiative heat losses provide precise flame structure information on radical species transport and reactions, which cannot be obtained by experiments or theories. Thus, this research contributes to expand our understanding of diffusion flame stabilization, particularly into a previously unstudied diffusion-dominant regime in microgravity. Furthermore, flame spread over condensed fuel surfaces and flame suppression have rarely been viewed from the point of stabilization and destabilization of edge diffusion flames. This research uniquely addresses to the edge diffusion flame phenomena in relation to the internal flame structure of the reaction kernel (peak reactivity spot) at the flame edge. The effects of various fire-extinguishing agents on the flame stability, spread, and suppression will be revealed on a step-by-step basis by systematic experiments and computations . Therefore, this research, which capitalizes the unique microgravity environment, benefits development efforts of fire-extinguishing systems to enable safe and productive human space explorations of the moon and Mars through fundamental understanding of the edge diffusion flame characteristics.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Underlying physics and chemistry of edge diffusion flames are relevant to the performance of Earth-bound combustion systems, including gas turbines, diesel engines, and industrial burners. Flame holding in combustors is necessary to achieve design performance and safe operation of the system. Therefore, this research is also beneficial for improvements and further developments of Earth-bound combustion technologies and certainly contributes to a progress in fundamental combustion science.
	The structure and stability of laminar jet diffusion flames of gaseous hydrocarbon fuels in co-flowing air at normal earth gravity have been investigated experimentally and computationally. Measurements of the critical mean jet velocity of methane, ethane, and propane at lifting or blowoff were made as a function of the air velocity. Computations using an implicit, third-order accurate numerical scheme simulated the internal structures of the stabilizing region (base) of methane and propane flames. A reaction mechanism, including 33 species and 112 elementary steps for C1-C3 hydrocarbons, and a broad-band radiation heat-loss model assuming optically thin media, were used. For methane and propane, the simulated flame base moved downstream toward lift-off under flow conditions close to the measured lifting limit. A peak reactivity spot (i.e., reaction kernel) formed in the stabilizing region is responsible for the flame attachment and detachment. The flame lift-off occurs due to flame-base destabilization (detachment), rather than extinction. An increase in the entrained velocity (decrease in the residence time) moves the flame base downstream, in search of a new balancing point between the residence time and overall reaction time, while still maintaining vigorous burning at the reaction kernel. SUPPLEMENTAL RESEARCH PROGRESS FOR FINAL REPORTING The structure, stabilization, and extinguishment of a low-speed co-flow diffusion flame in normal earth gravity (1g) and microgravity have been studied experimentally and computationally. In addition to a circular tube burner, a cup burner, which was used in a previous effort, and cylindrical burner geometries were used. For the tube and cup-burner configurations, methane, ethane, propane, or n-heptane was used as the fuel and an inert or chemically active gaseous agent was added to the coflowing air. The microgravity cup-burner flame extinguishment testing was conducted using

Task Progress:	standard and pre-extravehicular-activity (EVA) oxidizing atmospheres (21 % O2 in N2 at 101 kPa and 30 % O2 in N2 at 70.3 kPa, respectively) aboard the NASA Reduced-Gravity Aircraft (C-9) in March and October 2006. For the cylindrical configurations, three types of flames were used: (I) burning in a middle cylinder section, (II) around a downstream-end section, and (III) around an upstream-end section. Methane-blowing porous cylinders and cylindrical samples of polymers (polymethyl-methacrylate, high-density polyethylene, and polyoxymethylene) were used as the fuel source. A gaseous fire-extinguishing agent or water mist was added gradually to the oxidizer flow to measure the critical volume fraction of agent at extinguishment. The methane cylindrical flames extinguished via blowoff processes (in which the flame drifted downstream), whereas the polymer flames receded downstream and shrank, while the flame base kept attached to the surface, until extinguishment. The agent concentration in the oxidizer required for extinguishment was fairly independent of the mean oxidizer velocity (3 to 16 cm/s). By contrast, the water mist droplet mass fraction at extinguishment in cylindrical burners decreased substantially with increasing the mean oxidizer velocity. Numerical simulations with full chemistry were performed to reveal the detailed flame structure and to predict the minimum extinguishing concentrations (MECs) of gaseous agents for a comparison with the experiment. Despite the complexity of chemistry and flame-flow interactions in the blowoff processes, the numerical predictions of MECs were in fair agreements with the measurements. The MECs in microgravity were, in general, substantially larger than those in earth gravity, and the MECs in the pre-EVA condition were significantly larger than those in the standard atmosphere in microgravity. The relative ranking of agent effectiveness depended on the agent's ability to reduce the flame temperature to a threshold range through the heat capacity and thermal conductivity of the agent-laden oxidizer.
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